

TYPES AND λ -CALCULUS

Problem Sheet 2

Questions 1, 2 6 and 7 will be marked.

- * 1. Write these terms using the minimum number of parentheses and λ , according to our conventions.
- (a) $(\lambda y. ((y\ y)(z\ z)))$
 - (b) $(\lambda y. (((y\ y)\ y)\ y))$
 - (c) $((S\ Z)(\lambda y. (\lambda z. (z\ (S\ y))))))$
- * 2. Write the term $(\lambda x y z. x y (x z))(\lambda x y. x)$ with all the parentheses and λ , that we will usually omit, tediously put back in.
- * 3. List the all the subterms of the following terms (don't bother listing the same subterm more than once even it occurs several times):
- (a) The 3 distinct subterms of $\lambda x. x x$
 - (b) The 6 distinct subterms of $(\lambda x. x x)(\lambda y. y)$
 - (c) The 8 distinct subterms of $\lambda x y z. x y (y x)$
 - (d) The 13 distinct subterms of $\text{fix } (\lambda x y. \text{ifz } y\ y\ (x\ (\text{pred } y)))$
- * 4. Each of the following has two free variables, what are they in each case?
- (a) $\lambda x y. \lambda u. u v x y z$
 - (b) $\lambda x y. z(\lambda u. u v x y)$
 - (c) $\lambda w x. z(\lambda u. u v w x)$
 - (d) $\lambda v w. z(\lambda z. u v v w)$
 - (e) $\lambda y x. z(\lambda u. u w y x)$

* 5. Which of the following pairs of strings are α -equivalent (and therefore represent the same term):

- (a) $\lambda x. xy$ and $\lambda z. zy$
- (b) $\lambda x. xy$ and $\lambda z. zx$
- (c) $\text{ifz } x \text{ (S } x \text{) (pred } x \text{)}$ and $\text{ifz } y \text{ (S } y \text{) (pred } y \text{)}$
- (d) $\lambda xy. xy$ and $\lambda xy. yx$
- (e) $\text{fix } (\lambda x. (\lambda y. xy) \text{ (S } x \text{)})$ and $\text{fix } (\lambda y. (\lambda x. yx) \text{ (S } y \text{)})$

* 6. Perform the following substitutions:

- (a) $(\text{ifz } x \text{ (S } x \text{) Z})[\underline{2}/x]$
- (b) $\underline{2}[\underline{1}/x]$
- (c) $(\lambda x. (\lambda y. xz)z)[(\lambda z. z)/z]$
- (d) $(\lambda x. yx)[yz/x]$
- (e) $(\lambda x. yz)[yy/z]$
- (f) $(\lambda y. xy)[yx/x]$

** 7. Complete the following partial proof of the following statement, by filling out (a), (b) (c) and (d). This proof is a particularly good test of how carefully you are keeping track of the proof state.

Suppose P and Q are terms, x and y are variables. If $x \neq y$ and $x \notin \text{FV}(Q)$ then for all terms M :

$$M[P/x][Q/y] = M[Q/y][P[Q/y]/x]$$

(which tells us how to reorder substitutions in general).

Proof. Suppose P and Q are terms. Suppose x and y are variables. The rest of the proof is by induction on $M \in \Lambda$.

- In case M is some variable z , we argue as follows. Assume $x \neq y$ and $x \notin \text{FV}(Q)$. Our goal is to show:

$$z[P/x][Q/y] = z[Q/y][P[Q/y]/x]$$

Now, either $z = x$, $z = y$ or z is neither x nor y . We proceed by a case analysis on this fact:

- Suppose $z = x$. By our assumption, it follows that $z \neq y$. Then, by definition of substitution, $z[P/x][Q/y] = P[Q/y]$ and

$$z[Q/y][P[Q/y]/x] = z[P[Q/y]/x] = P[Q/y]$$

Hence, the desired equality follows.

- Suppose $z = y$. (a)
- Suppose $z \neq x$ and $z \neq y$. Then $z[P/x][Q/y] = z$ on the left side of the goal and also $z[Q/y][P[Q/y]/x] = z$ on the right side, so the result follows.
- (b)

- In case M is some application N_1N_2 we argue as follows. Assume $x \neq y$ and $x \notin \text{FV}(Q)$. Additionally, assume the induction hypothesis:

(IH1) if $x \neq y$ and $x \notin \text{FV}(Q)$ then $N_1[P/x][Q/y] = N_1[Q/y][P[Q/y]/x]$

(IH2) if $x \neq y$ and $x \notin \text{FV}(Q)$ then $N_2[P/x][Q/y] = N_2[Q/y][P[Q/y]/x]$

Our goal is to show that:

$$(N_1N_2)[P/x][Q/y] = (N_1N_2)[Q/y][P[Q/y]/x]$$

(c)

- In case M is of shape $\lambda z.N$, we argue as follows. We can assume by the bound variable convention that z is different from any variable in scope, so $z \neq x$ and $z \neq y$ and $z \notin \text{FV}(P)$ and $z \notin \text{FV}(Q)$. (d)

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